



PowerPoint Presentation by
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MANAGEMENT ACCOUNTING

8th EDITION

BY

HANSEN & MOWEN

12 TACTICAL DECISION MAKING

LEARNING OBJECTIVES

After studying this chapter, you should be able to:

LEARNING OBJECTIVES

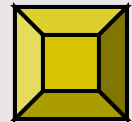
1. Describe the tactical decision-making model.
2. Explain how the activity resource usage model is used in assessing relevancy.
3. Apply tactical decision-making concepts in a variety of business situations.

Continued

LEARNING OBJECTIVES

4. Choose the optimal product mix when faced with one constrained resource.
5. Explain the impact of cost on pricing decisions.
6. Use linear programming to find the optimal solution to a problem of multiple constrained resources. (*Appendix*)

*Click the button to skip
Questions to Think About*



QUESTIONS TO THINK ABOUT: Tidwell Products, Inc.

Describe the decision to be made by Tidwell. Is it a strategic or tactical decision?

QUESTIONS TO THINK ABOUT: Tidwell Products, Inc.

What costs do you think Leo is referring to in the last paragraph of the scenario? Give examples?

QUESTIONS TO THINK ABOUT: Tidwell Products, Inc.

Assume Tidwell Products accepts Linda's first alternative. Are there any noncost factors that should be considered? What about her second alternative?

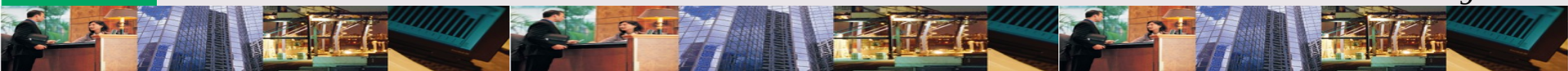
LEARNING OBJECTIVE

1

Describe the tactical decision-making model.

Is there a difference
between tactical and
strategic decisions?

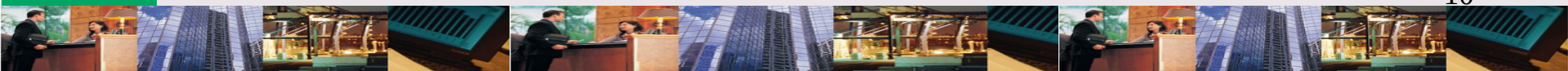
Yes! Tactical & strategic
decisions differ on the time
period affected.



TACTICAL DECISION MAKING:

Definition

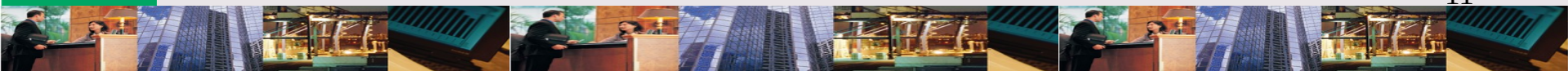
Consists of choosing among alternatives with an **immediate or limited end** in view.



STRATEGIC DECISION MAKING:

Definition

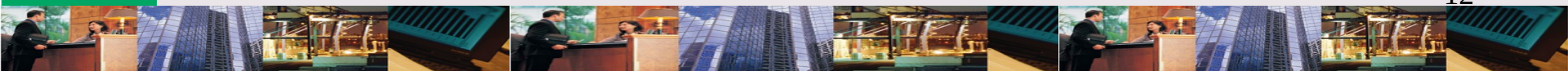
Is selecting among alternative strategies so that long term competitive advantage is established.



TACTICAL MODEL

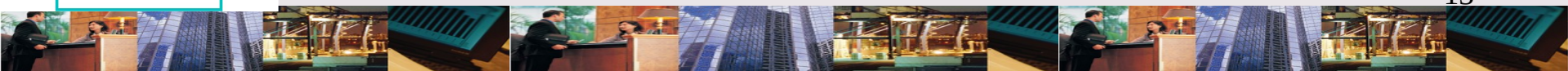
A general approach to tactical decision making includes:

1. Recognize, define the problem
2. Identify alternatives, eliminating those that are unfeasible
3. Identify costs & benefits
4. Total relevant costs, benefits of each alternative
5. Assess qualitative factors
6. Select alternative with greatest overall benefit



TIDWELL PRODUCTS: Background

Tidwell Products Inc. is facing expanded production that is straining the capacity in facilities with 5 years remaining on their lease. Two feasible alternatives under consideration are **a)** to rent an additional building for warehousing and **b)** outsource production. The CFO will prepare a report of detailed costs for these alternatives.



APPLYING TACTICAL MODEL

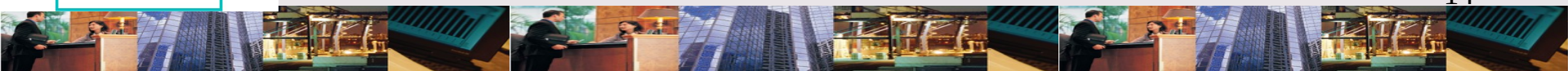
Step 1: Define the problem

Increase capacity for warehousing & production

Step 2: Identify alternatives

1. Build new facility
2. Lease larger facility; sublease current facility
3. Lease additional facility
4. Lease warehouse space
5. Buy shafts & bushings; free up space

Continued



APPLYING TACTICAL MODEL

Step 3: Identify costs, benefits

Alt 4: $\langle \text{Costs} \rangle + \text{Benefits}$

Step 4: Total relevant costs & benefits

Alt 5: $\langle \text{Costs} \rangle + \text{Benefits}$

Alt 4: Relevant $\langle \text{Costs} \rangle + \text{Benefits}$

Alt 5: Relevant $\langle \text{Costs} \rangle + \text{Benefits}$

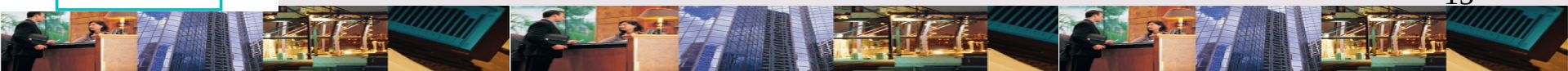
Step 5: Assess qualitative factors

Differential cost

1. Quality of external supplier
2. Reliability of external supplier
3. Price stability
4. Labor relations & community image

Step 6: Make decision

Continue producing & lease warehouse



TIDWELL'S TACTICAL MODEL: Detailed Costs

Tidwell Productions estimates the following costs for feasible alternatives #4 & #5 are equal:

Alt. 4:

Variable costs	\$ 345,000
Warehouse lease	135,000

Alt. 5:

Purchase price	\$ 460,000
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Continued

TIDWELL'S TACTICAL MODEL: Detailed Costs

Tidwell Productions estimates the following **relevant total costs** for feasible alternatives #4 & #5 are different:

Alt. 4: \$ 480,000

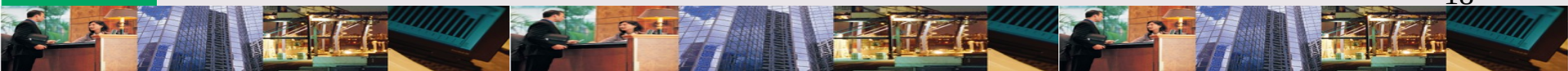
Alt. 5: 460,000

Differential cost \$ 20,000

Although costs of Alternative #4 exceed the costs of Alternative #5, **qualitative factors outweigh cost** concerns. Tidwell should lease the warehouse & produce shafts & bushing internally.

RELEVANT COSTS: Definition

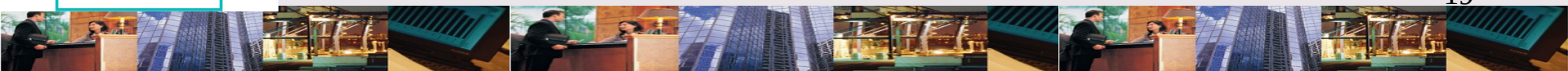
Are future costs that differ across alternatives.



RELEVANT COSTS ILLUSTRATED

In Tidwell Products' decision, the cost of direct labor (\$150,000 of variable production costs) is a relevant cost because it differs between Alternatives #4 & #5.

There is no labor cost if shafts & bushings are purchased externally.



IRRELEVANT COSTS ILLUSTRATED

In Tidwell Products' decision, the depreciation cost of the leased building is irrelevant because it is a sunk cost that

- a) Is not affected by future actions;
- b) Can not be avoided; and
- c) Does not differ across alternatives.

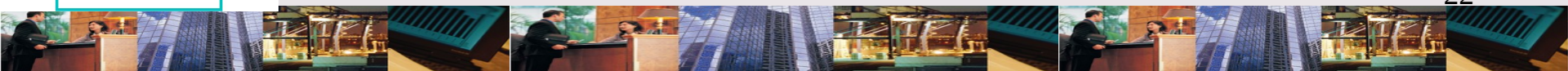
RELEVANT VS. IRRELEVANT COSTS

	Cost to Make	Cost Not to Make	Differential Cost
Direct labor	\$ 150,000	---	\$ 150,000
Depreciation	125,000	\$ 125,000	---
Allocated lease	<u>12,000</u>	<u>12,000</u>	<u>---</u>
	<u>\$ 287,000</u>	<u>\$ 137,000</u>	<u>\$150,000</u>

Direct labor is the relevant cost because it differs between alternatives.

TACTICAL DECISIONS & ETHICS

If Tidwell were to lay off workers for tactical advantage that did not support long term goals, ethics of the decision would be questionable.



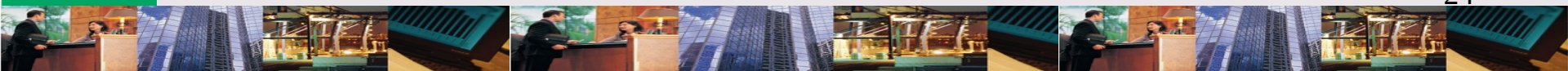
LEARNING OBJECTIVE

2

Explain how the activity resource usage model is used in assessing relevancy.

How can the activity
resource usage model be
used to assess relevance?

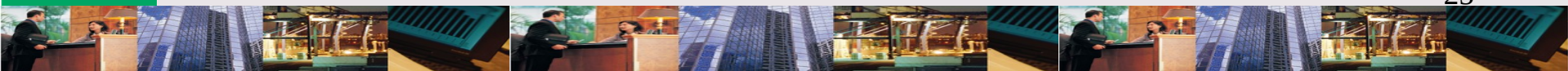
To assess relevance, resources
must be identified as flexible or
committed.



FLEXIBLE RESOURCES:

Definition

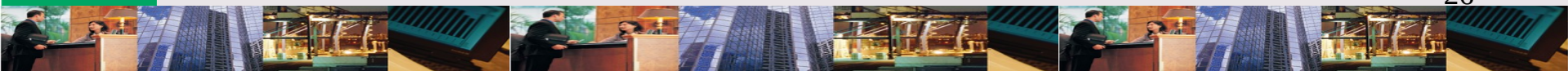
Are **a)** easily purchased in the amount needed **b)** at the time of use.



COMMITTED RESOURCES:

Definition

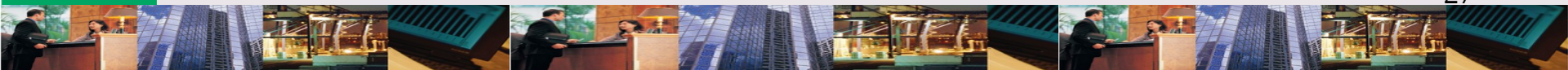
Are **a)** purchased before they are needed & **b)** may not be completely used.



MANUFACTURING FIRM:

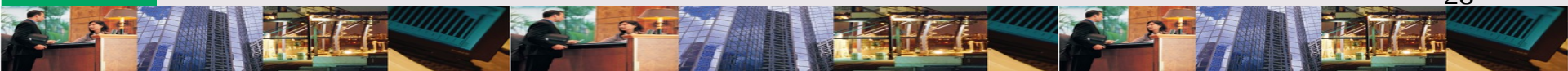
Background

A manufacturing firm employs five (5) engineers with a capacity of 10,000 engineering hours (2,000 hours each) at a cost of \$250,000 (\$25 per hour). The firm expects to use only 9,000 engineering hours during the current year, producing unused capacity.



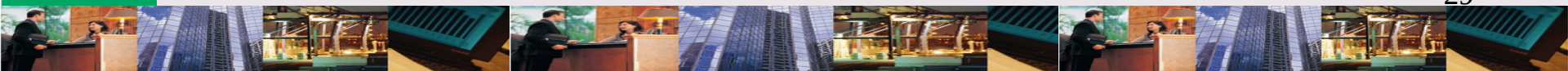
Should the firm consider accepting a special order that uses 500 engineering hours?

Yes. The firm should consider accepting the special order, if it is otherwise profitable, because it will be completed with unused engineering capacity.



Would circumstances be different if the special order uses 1,500 engineering hours?

Yes. Since 1,500 exceeds available hours of engineering labor, the company must weigh the cost of additional hiring or consulting against the gain in profit.



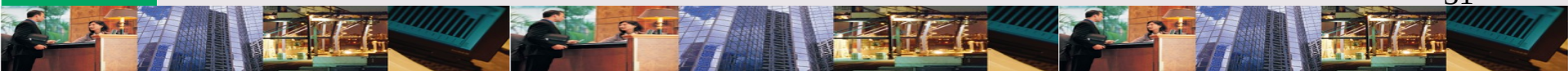
LEARNING OBJECTIVE

3

Apply tactical decision-making concepts in a variety of business situations.

TACTICAL DECISION- MAKING: Examples

- ✧ Make-or-Buy Decisions
- ✧ Keep or Drop
- ✧ Keep or Drop & Replace
- ✧ Special order
- ✧ Sell or process further

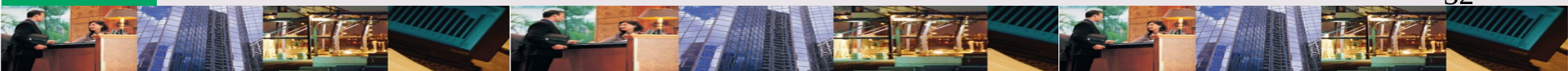


SWASEY MANUFACTURING :

Make-or-Buy Background

Swasey Manufacturing, a printer manufacturer, will switch to a printer that does not use an electronic component it currently produces. Should Swasey produce 10,000 components for the older printer this year or should they purchase the component for \$4.75?

Continued



SWASEY MANUFACTURING :

Make-or-Buy Background

	Total Cost	Unit Cost
Equipment Rent	\$ 12,000	\$ 1.20
Equipment depreciation	2,000	0.20
Direct materials	10,000	1.00
Direct labor	20,000	2.00
Variable overhead	8,000	0.80
General fixed overhead	<u>30,000</u>	<u>3.00</u>
Total	<u>\$ 82,000</u>	<u>\$ 8.20</u>

**Unit costs are
calculated on the
basis of producing
10,000 printers.**

Continued

SWASEY'S TACTICAL MODEL: Make-or-Buy

Step 1: Define the problem

Have component available for old printer

Step 2: Identify alternatives

1. Make component
2. Buy component

Step 3: Identify costs, benefits

1. Make: \$8.20
2. Buy: \$475

Step 4: Total relevant costs & benefits

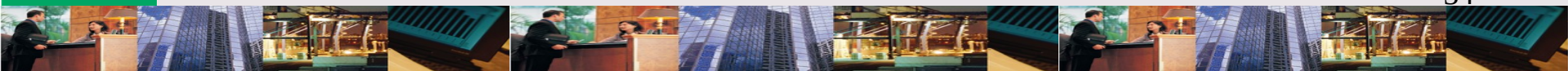
Omit depreciation & allocated fixed factory overhead.

Step 5: Assess qualitative factors

?

Step 6: Make decision

?



SWASEY MANUFACTURING:

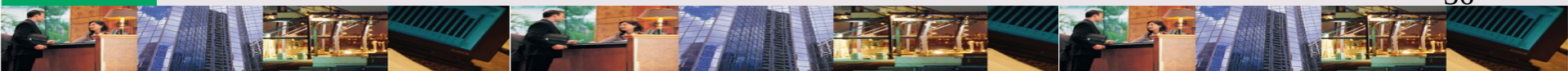
Relevant Information

	Alternatives		Differential
	Make	Buy	Cost to Make
Equipment Rent	\$ 12,000	---	\$ 12,000
Direct materials	5,000	---	5,000
Direct labor	20,000	---	20,000
Variable overhead	8,000	---	8,000
Purchased cost	---	\$ 47,500	(47,500)
Receiving Dept labor	<u>---</u>	<u>8,500</u>	<u>(8,500)</u>
Total	<u>\$ 45,000</u>	<u>\$ 56,000</u>	<u>\$ (11,000)</u>

SWASEY MANUFACTURING:

Make-or-Buy Analysis

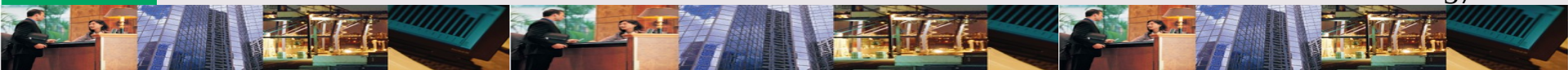
Because Swasey Manufacturing must hire labor to staff the Receiving department, buying the component will cost **\$5.60** per unit. Swasey should produce the component because the component requires **\$4.50** in relevant production costs per unit.



NORTON MATERIALS: Keep-or-Drop Background

Norton Materials produces 3 products: blocks, bricks, and tile. The tile segment has a negative segment margin and does not contribute to common fixed expenses. Should Norton drop the tile division?

Continued



NORTON MATERIALS: Keep-or-Drop

	Blocks	Bricks	Tiles	Total
Sales	\$ 500	\$ 800	\$ 150	\$ 1,450
Less Variable exp.	<u>250</u>	<u>480</u>	<u>140</u>	<u>870</u>
Contribution margin	\$ 250	\$ 320	\$ 10	\$ 580
Less direct fixed exp				
Advertising	\$ 10	\$ 10	\$ 10	\$ 30
Salaries	37	40	35	112
Depreciation	53	40	10	103
Total	<u>\$ 100</u>	<u>\$ 90</u>	<u>\$ 55</u>	<u>\$ 245</u>
Segment margin	<u>\$ 150</u>	<u>\$ 230</u>	<u>\$ (45)</u>	\$ 335
Less Common fixed exp				<u>125</u>
Operating income				<u>\$ 210</u>

Continued

NORTON'S TACTICAL MODEL: Make-or-Buy

Step 1: Define the problem

Tile division does not contribute to common fixed expenses

Step 2: Identify alternatives

1. Keep division
2. Drop division

Step 3: Identify costs, benefits

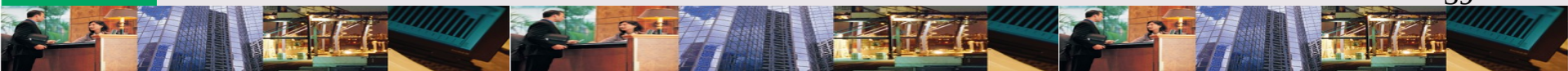
1. Keep: saves \$10,000 CM
2. Drop: eliminates \$45,000 segment loss

Step 4: Total relevant costs & benefits

Should loss of other sales be considered?

Step 5: Assess qualitative factors

Step 6: Make decision



NORTON: President's Analysis (000)

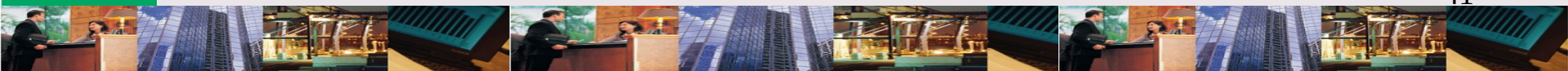
	Keep	Drop	Keep Difference
Sales	\$ 150	---	\$ 150
Less Variable exp.	140	---	140
Contribution margin	\$ 10	---	\$ 10
Less Advertising exp	(10)	---	(10)
Cost of supervision	<u>(35)</u>	---	<u>(35)</u>
Total benefit (loss)	<u>(35)</u>	---	<u>(35)</u>

President's analysis suggests that Tile should be dropped.

Continued

Can the Tile Division be dropped with no effect on other divisions?

No. Dropping tiles will decrease sales of both blocks and bricks.



NORTON: Marketing Perspective (000s)

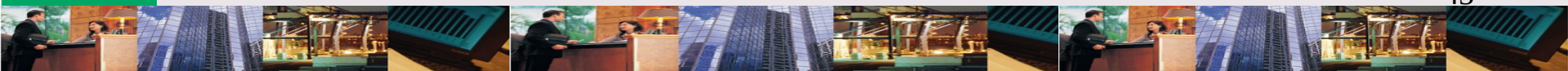
	Keep	Drop	Keep Difference
Sales	\$ 1,450	\$ 1,186.0	\$ 264.0
Less Variable exp.	<u>870</u>	<u>666.6</u>	<u>203.4</u>
Contribution margin	\$ 580	\$ 519.4	\$ 60.6
Less Advertising exp	(30)	(20.0)	(10)
Cost of supervision	<u>(112)</u>	<u>(77.0)</u>	<u>(35)</u>
Total benefit (loss)	\$ <u>438</u>	\$ <u>422.4</u>	\$ <u>15.6</u>

Marketing's analysis suggests that Tile should be kept.

Continued

Can the Tile Division be changed to produce floor tile for a profit?

Yes. However it might not be as profitable as the current product mix.



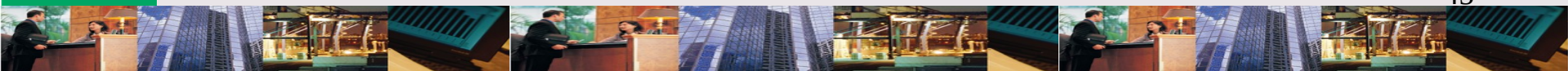
NORTON: Production Perspective (000s)

	Keep	Drop & Replace	Keep Difference
Sales	\$ 1,450	\$ 1,286.0	\$ 264.0
Less Variable exp.	<u>870</u>	<u>706.6</u>	<u>203.4</u>
Contribution margin	\$ <u>580</u>	\$ <u>579.4</u>	<u>\$ 0.6</u>

Production's replacement suggestion is not as profitable as keeping ceiling tiles.

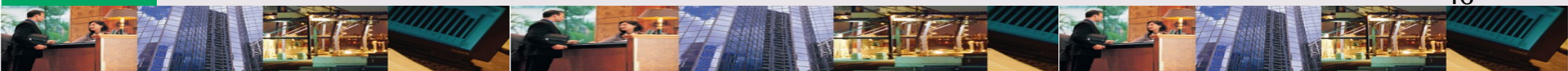
NORTON MATERIALS : Keep or Drop Analysis

Because Norton will lose sales in both blocks and brick if ceiling tiles are dropped and replacing ceiling tiles with floor tiles is less profitable, the firm is better off to keep the ceiling tile division.



SPECIAL ORDER: Definition

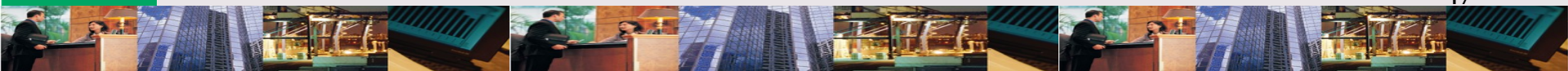
Decisions focus on whether a specially priced order should be accepted or rejected.



ICE CREAM: Special Order Background

An ice cream company is operating at 80% of its 20 million gallon capacity. The company receives an offer to purchase 2 million gallons for \$1.55 per gallon. This is below the wholesale price of \$2.00. Should the company accept the offer?

Continued



ICE CREAM TACTICAL MODEL: Special Order

Step 1: Define the problem

Is a special order profitable with excess capacity?

Step 2: Identify alternatives

1. Accept
2. Reject

Step 3: Identify costs, benefits

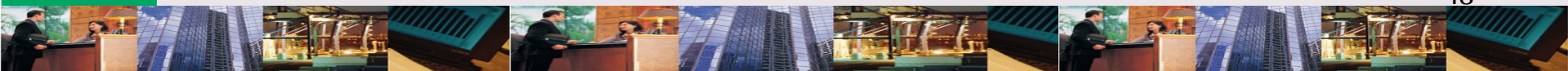
With excess capacity, opportunity for profit

Step 4: Total relevant costs & benefits

Will the price cover variable product costs

Step 5: Assess qualitative factors

Step 6: Make decision



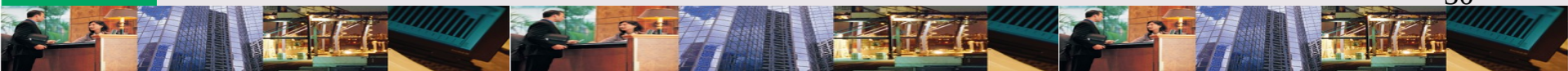
ICE CREAM: Special Order (000s)

	Accept	Reject	Benefit Difference
Sales	\$ 3,100	---	\$ 3,100
Dairy ingredients	(1,400)	---	(1,400)
Sugar	(200)	---	(200)
Flavoring	(300)	---	(300)
Direct labor	(500)	---	(500)
Packaging	<u>(400)</u>	<u>---</u>	<u>(400)</u>
Other	<u>(100)</u>	<u>---</u>	<u>(100)</u>
Profit	<u>\$ 200</u>	<u>---</u>	<u>\$ 200</u>

Using relevant information, the special order adds \$200,000 to profit.

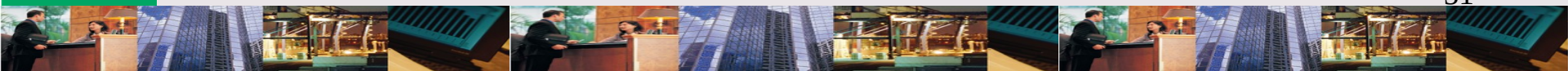
ICE CREAM : Special Order Analysis

Even though the special order price for 2 million gallons of ice cream is below the normal selling price of \$2.00, it will be profitable because there is spare capacity and only relevant variable costs are considered in the decision.



JOINT PRODUCTS: Definition

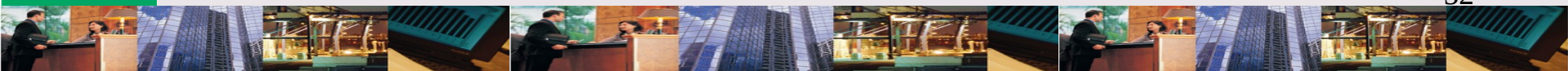
Have common processes &
cost of production up to a
split-off point.



APPLETIME: Sell or Process Background

Appletime grows and sells apples in grades A, B, & C. Grade B apples are usually bagged & sold. However, a supermarket is offering to buy apple pie filling that Appletime would make from grade B apples. Should Appletime process grade B apples into apple pie filling?

Continued



APPLETIME JOINT PRODUCTION

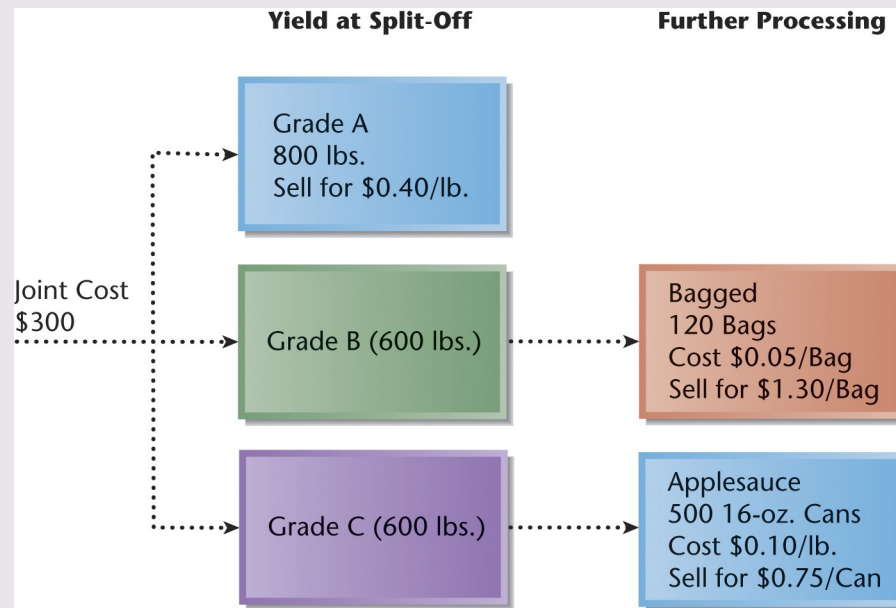


EXHIBIT 12-3

APPLETIME TACTICAL MODEL: Process Further

Step 1: Define the problem

Will it be profitable to process grade B apples further?

Step 2: Identify alternatives

1. Accept
2. Reject

Step 3: Identify costs, benefits

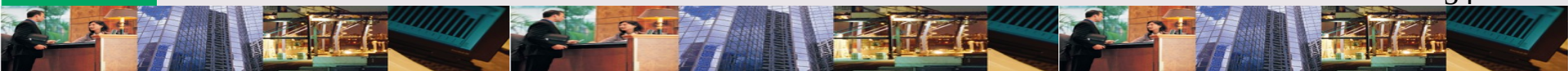
Weigh processing costs against selling price

Step 4: Total relevant costs & benefits

Is there more profit in processing further?

Step 5: Assess qualitative factors

Step 6: Make decision



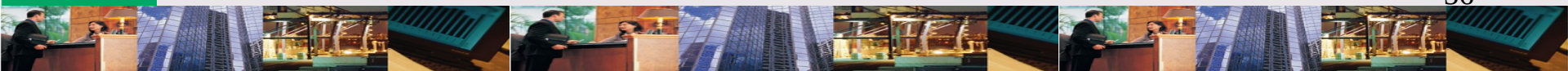
APPLETIME: Process Further

	Process	Sell	Process Difference
Sales	\$ 450	\$ 150	\$ 300
Processing cost	<u>120</u>	<u>---</u>	<u>120</u>
Total	\$ <u>330</u>	\$ <u>150</u>	<u>\$ 180</u>

By processing grade B apples into pie filling, profit will increase.

APPLETIME : Process Further Analysis

Even though processing grade B apples further increases costs, there is more profit to be made from making pie filling than from selling grade B apples by the bag.



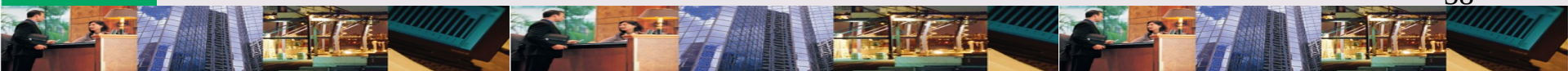
LEARNING OBJECTIVE

4

Choose the optimal product mix when faced with one constrained resource.

CONSTRAINTS: Definition

Are limitations a business faces such as limited resources or demand.



LEARNING OBJECTIVE

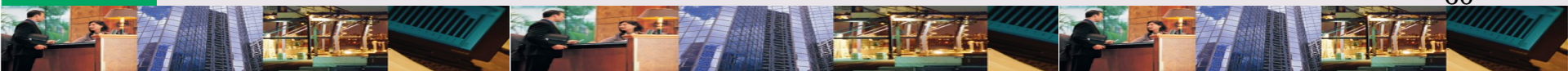
5

Explain the impact of cost on pricing decisions.

COST-BASED PRICING:

Definition

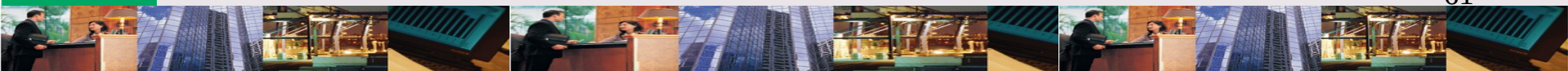
Means setting a sales price based on marking up a base cost such as COGS or direct materials by a certain percentage.



TARGET COSTING & PRICING:

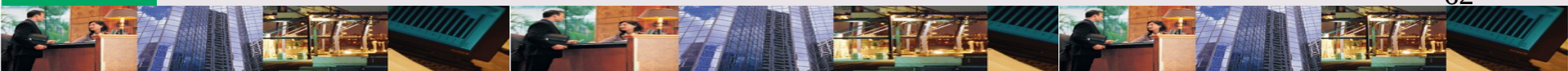
Definition

Is price-driven costing, i.e., deriving cost based on setting a target price that customers are willing to pay for the good or service.



PRICING: Legal Aspects

- ✧ Predatory pricing
 - ✧ A means of setting price to eliminate competition
 - ✧ Dumping on international market
- ✧ Price discrimination
 - ✧ Charging different prices to different customers
- ✧ Price gouging
 - ✧ Using market power to set prices too high



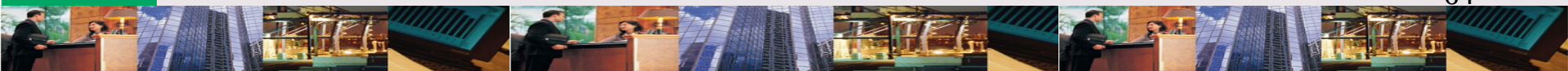
LEARNING OBJECTIVE

6

Use linear programming to find the optimal solution to a problem of multiple constrained resources. (*Appendix*)

LINEAR PROGRAMMING: Definition

Is a mathematical method of finding an optimal solution to a production problem.



GRAPHING SOLUTION

Linear programming demonstrates the feasible production region & optimal solution for complex problems.

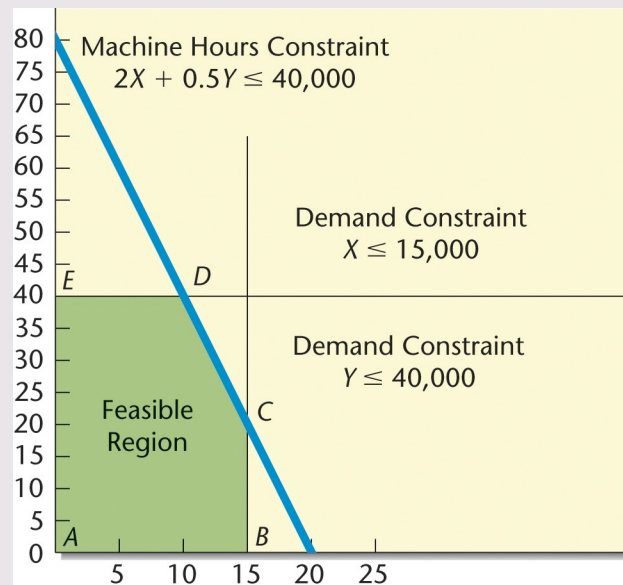


EXHIBIT 12-4

CHAPTER 12

THE END